SCIT-EIS-UOW
CSCI291 Programming for Engineers
Autumn 2024
Assignment 1 - Due date: April 19 2025 at 11:55 pm

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General code notes

These are some general rules about what you should and shouldn't do.

1. Your code must compile without warnings or errors on the command line in the terminal with the command:

   ```
   gcc -std=c11 precision_farm.c -Wall -lm -o precision_farm
   ```

2. Your code must be executable as:

   ```
   precision_farm number_of_days number_of_plots(maximum 20) resource_cost.dat
   ```

3. The structure of the text file `resource_cost.dat` is as follows (per line):

   Water: 0.002

   Fertilizer: 0.5

   Pesticide: 5.00

   Wheat: 0.30; Corn: 0.20; Rice: 0.40; Soybean: 0.60;

   and must be read from within your program.

4. Do not hardcode these arguments in your code because the values can be changed and values in the cost f resources can change.

5. Within your code, be consistent in your tabbing style. We want readable code.

6. Include sensible volumes of commenting otherwise you will lose marks.

7. Use functions to divide the functionality of your code into sensible components.

8. You will lose a lot of marks for not using functions.

9. Use appropriate and sensible variable names.

10. Don't leak memory.

Description

The aim of this assignment is to provide opportunity for students to develop and exercise their knowledge of programming concepts in designing software that solves a simple near-real-life problem. Such concepts include:

1. Data types (basic, enum, struct, union), expressions, files and operators
2. Loops, selection and functions
3. Standard math functions, random number generation
4. Pointers, arrays and memory management

Specifications

The purpose of this assignment is to simulate the management of a hypothetical farm with fields of crops. Your program will use random number generator to simulate environmental factors and implement algorithms to make efficient farming decisions based on current conditions and weather forecast. The simulation is to be written in the `C` programming language. Ensure that you break the logic of your code into functions callable from `main`. Your `main` will take arguments from the command line as sated above uner "General code notes". Broadly speaking the narrative is as follows. You are an engineer employed in a precision farming company that deploys sensors to monitor and maintain a farm of crops. The objective is to make a profit from this business. Hence, the crops must be monitored and maintained to maximise the yield. The farm specialises in wheat, corn, rice and soybean, where average yield per hectare of the respective crops are : wheat - 2,880 kg/ha; corn - 10,000 kg/ha; rice - 4,500 kg/ha; soybean - 2,000 kg/ha (values obtained from Australian agricultural data). Each crop has a maturity period: wheat - 120 days; corn 100 days; rice 150 days; soybean 110 days. If the simulation runs for fewer days than maturity, yield is scaled proportionally to the maturity ratio. For each crop, the plot will have the following attributes:

- Crop type (use an enum for Wheat, Corn, Rice, Soybean)
- Number of plots per crop (randomized, up to 4 per crop)
- Soil moisture level
- Nutrient levels: Nitrogen (N), Phosphorus (P), Potassium (K), Iron (Fe)
- Pest probability and pest damage (i.e if probability is above a threshold assign a level of damage). Use sensible values.
- Rain forecast; Amount of rain determines moisture level of the plot.
- Number of days grown
- Resource usage: water, fertilizer, pesticide
- Final yield and profit/loss calculation

1. Use random number generation (we have used this code in the lab practice) to simulate:
    - Initial values for nutrients, moisture, pest probability, and rain forecast
    - Daily environmental fluctuations (moisture loss, rainfall)
    - Pest events and progression
    - Random assignment of 1-4 plots per crop

2. Nutrient Monitoring and Depletion:
   - Nutrient levels (N, P, K, Fe) should decrease each day following an exponential decay model: nutrient_today = nutrient_yesterday × decay_factor (try 0.99)

     If any nutrient falls below a critical threshold (try 30%), fertilizer must be applied.
   - Fertilizer restores all nutrients partially (try +20 units per application).
3. Rain forecast and irrigation:
   - At the beginning of each day, the soil loses a random amount of moisture due to environmental evaporation. This is between 5 and 10 units,
   - If it is forecast that it will rain (i.e rain_forecast == 1), a random amount of moisture is added to the soil, between 5 and 15 units.
4. Decision Logic:
   - Irrigate when moisture is less than a threshold (try 30%).
   - If irrigation is needed, water is added to bring it back to the threshold. water_needed = (MOISTURE_THRESHOLD - plot_moisture) x 10
   - Fertilize when any nutrient is deficient.
   - Apply pesticide when pests are detected.
   - Adjust final yield based on:
     - Nutrient sufficiency (each nutrient below threshold reduces yield by 10%)
   - Pest damage (reduces yield by 5% per damage level)
   - Maturity ratio is (days_grown / maturity_days)
5. Economic Simulation:
   - Resource Costs (to be read from a file called `resource_cost.dat`). So do not hardcode this into your program as they can change in the file used to test your code.:
     - Water: $0.002 per litre
     - Fertilizer: $0.5 per kg
     - Pesticide: $5.00 per litre
     - Market Prices:
       * - Wheat: $0.30/kg, Corn: $0.20/kg, Rice: $0.40/kg, Soybean: $0.60/kg
     - Revenue = Yield × Market Price
     - Profit/Loss = Revenue - Total Resource Cost
6. Output to display:
   - Crop type, number of plots, days simulated vs. maturity, maturity ratio, final yield, total water/fertilizer/pesticide used, total cost, revenue, and profit/loss.
   - Clearly identify any plots that are unprofitable (i.e., profit ¡ 0). Example output is given next.
7. Sample output:

   Plot 1 - Crop: Wheat

   Days Grown: 90 / 120

   Yield: 1987.45 kg

Price: \$0.30/kg

Revenue: \$596.24

Expenditure: \$115.00

Water Used: 500.00 L

Fertilizer Used: 10.00 kg

Pesticide Used: 1.00 L

Profit/Loss: \$481.24

Plot 2 - Crop: Corn

Days Grown: 90 / 100

Yield: 4210.10 kg

Selling Price: \$0.20/kg

Revenue: \$842.02

Expenditure: \$132.00

Water Used: 400.00 L

Fertilizer Used: 12.00 kg

Pesticide Used: 2.00 L

Profit/Loss: \$710.02

Plot 3 - Crop: Soybean

Days Grown: 90 / 110

Yield: 1020.30 kg

Selling Price: \$0.60/kg

Revenue: \$612.18

Expenditure: \$195.50

Water Used: 800.00 L

Fertilizer Used: 15.00 kg

Pesticide Used: 3.50 L

Profit/Loss: \$416.68

Plot 4 - Crop: Rice

Days Grown: 90 / 150

Yield: 1720.00 kg

Selling Price: \$0.40/kg

Revenue: \$688.00

Expenditure: \$701.00

Water Used: 650.00 L

Fertilizer Used: 20.00 kg

Pesticide Used: 6.00 L

Profit/Loss: -\$13.00

>> This plot was unprofitable.